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Wiping off oxygen bonding to maximize heteroatom-induced improvement in oxygen reaction activity of metal site for high-performance zinc-air battery

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Abstract

Heteroatom doping has recently been utilized to improve the catalytic performance of transition metal-based electrocatalysts. However, the doping process is inevitably accompanied by the introduction of oxygen, influencing the heteroatom-induced asymmetric spin density over the active sites and leading to inconspicuous promotion in the property. Herein, by wiping off the undesired heteroatom-oxygen bonding, we maximize the heteroatom-induced improvement in oxygen reaction activity of metal site, providing descendant energy barrier and favorable reaction efficiency for zinc-air batteries. The proof-of-concept material delivers a superior half-wave potential of 0.88 V versus reversible hydrogen electrode for oxygen reduction reaction, a small overpotential of 410 mV at the current density of 10 mA cm⁻² for oxygen evolution reaction, and a reversible oxygen electrode index of 0.76 V in electrochemical measurements. Aqueous zinc-air battery with such catalysts delivers an excellent power density of 162.3 mW cm⁻² and superior durability over 635 cycles. Moreover, in consideration of high safety and flexibility of solid-state batteries, all-solid-state zinc-air battery adopting gel electrolyte is assembled and used to illumine an LED wristband, showing great promises for the next-generation energy system.

Supplementary material for this article is available [online](#)

Keywords: oxygen reduction reaction, heteroatom doping, zinc-air battery, all-solid-state battery

(Some figures may appear in colour only in the online journal)

Introduction

As energy sources are exploited by humankind on a large scale, energy crisis is one of the heated topics that people around the world are concerning, thus clean secondary cells are studied by increasing researchers and are going to play a significant role in leading a trend to portable electronic devices. Among various batteries, zinc-air batteries are viewed as the next-generation

energy systems owing to their high specific capacitance, high energy density, long storing life, low cost and environmental friendliness [1, 2]. Active substance oxygen in the cathode administers to reach a higher theoretical energy density when compared to other conventional batteries [3]. In spite of these benefits, large overpotentials are often observed in charging–discharging curves because of the sluggish kinetics of oxygen reduction/evolution reaction (ORR/OER) [4, 5].

Cathode electrocatalysts with sophisticated multi-electron transfer processes in electrode reactions can enhance

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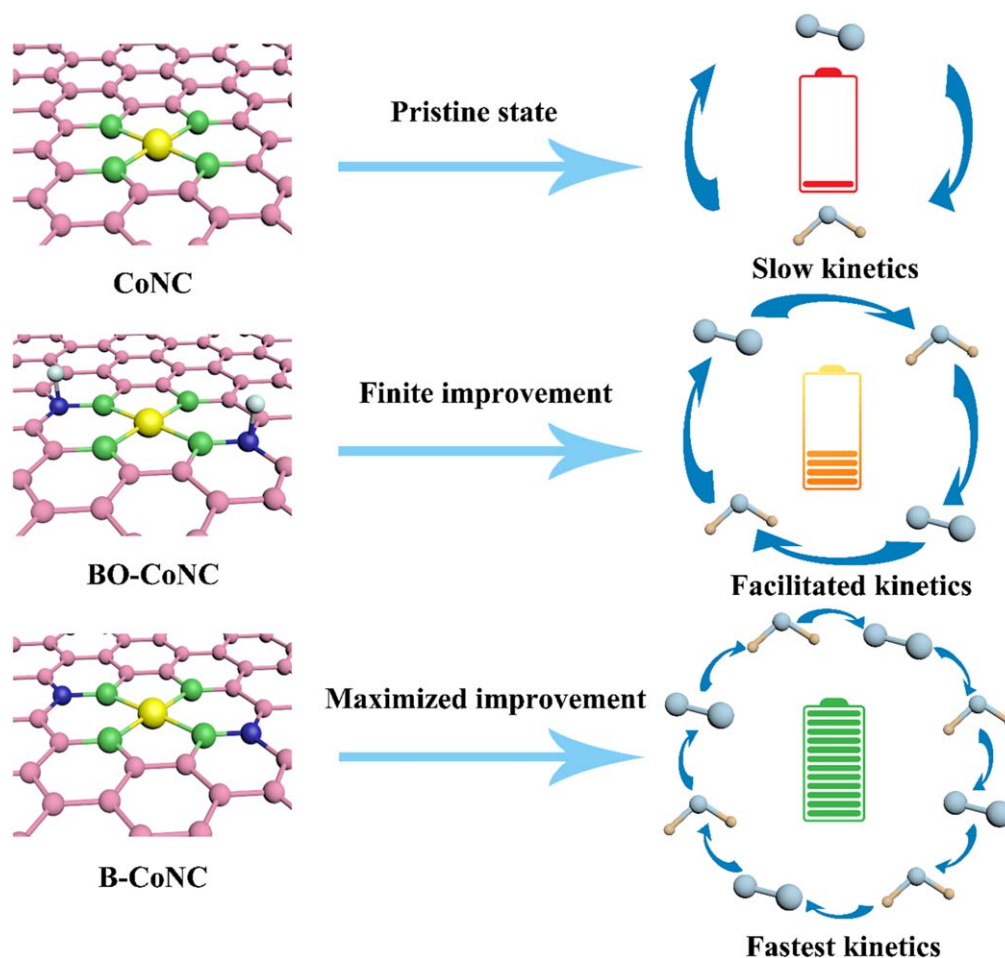


Figure 1. The molecular and reaction diagram of CoNC, BO-CoNC and B-CoNC. The pink, green, cyan, yellow, blue, and brown spheres represent C, N, O, Co, B, and H atoms, respectively.

oxygen reaction activity because of the descendant energy barrier and favorable reaction efficiency. Till now, some noble metals (such as Pt for the ORR and IrO₂ for the OER) have been common active catalysts in zinc-air batteries. Nevertheless, in the light of the high cost, scarcity and inferior stability of noble metal, the large-scale commercialization of zinc-air batteries is obstructed [6, 7]. Researchers have considered transition metal-based materials and carbon materials doped with heteroatom, including nitrogen, sulfur, boron and phosphorous, as the most prosperous catalysts to replace noble metal-based materials [8–11]. In particular, the introduction of transition metal atoms to nitrogen-doped carbon materials makes up metal-N-C moieties, which is extremely likely to substitute noble metal catalysts. To date, zeolitic-imidazolate-framework (ZIFs) with nitrogen-containing organic functional groups and transition metal ions, such as Co, have been extensively focused as precursors for fabricating CoNC catalytic materials, attributing to their high surface areas, optimized conductivity and uniform Co, N-doping after the pyrolysis process [12, 13]. Additionally, specific Co-based active sites could tune the electron structure and accelerate the adsorption of intermediates [14]. Despite recent advances, the catalytic performance of pure Co active sites, however, is still under satisfactory. Investigators further

discover that the doping heteroatoms like boron atoms [15] into Co-containing materials may benefit the activity, nonetheless B–O bonding are often inevitably introduced into such materials. With such undesired O-functionalities on the catalyst surface, several crucial steps of the oxygen reaction, including intermediates adsorption and electron charge transfer, would be disturbed [16], leading to inconspicuous promotion in the reaction activity. What we are committed to is to avert the effects of B–O structure.

Herein, we demonstrate that, pure B doping without B–O bonding in CoNC structure could indeed maximizing the heteroatom's contribution in improving the oxygen reaction activity, thus boosting cathodic kinetics of zinc-air battery (figure 1). Without the influence of B–O bonding, the boron-induced asymmetric spin density over the active sites would be maximized, producing favorable reaction environment and promoting the bifunctional activity. As expected, the proof-of-concept material delivers a superior half-wave potential of 0.88 V versus reversible hydrogen electrode (versus RHE) for ORR, a small overpotential of 410 mV at the current density of 10 mA cm^{−2} for OER, and a reversible oxygen electrode index of 0.76 V in electrochemical measurements. With the help of these electrocatalysts, we abate the overpotentials of zinc-air batteries significantly, thus leading to a maximum

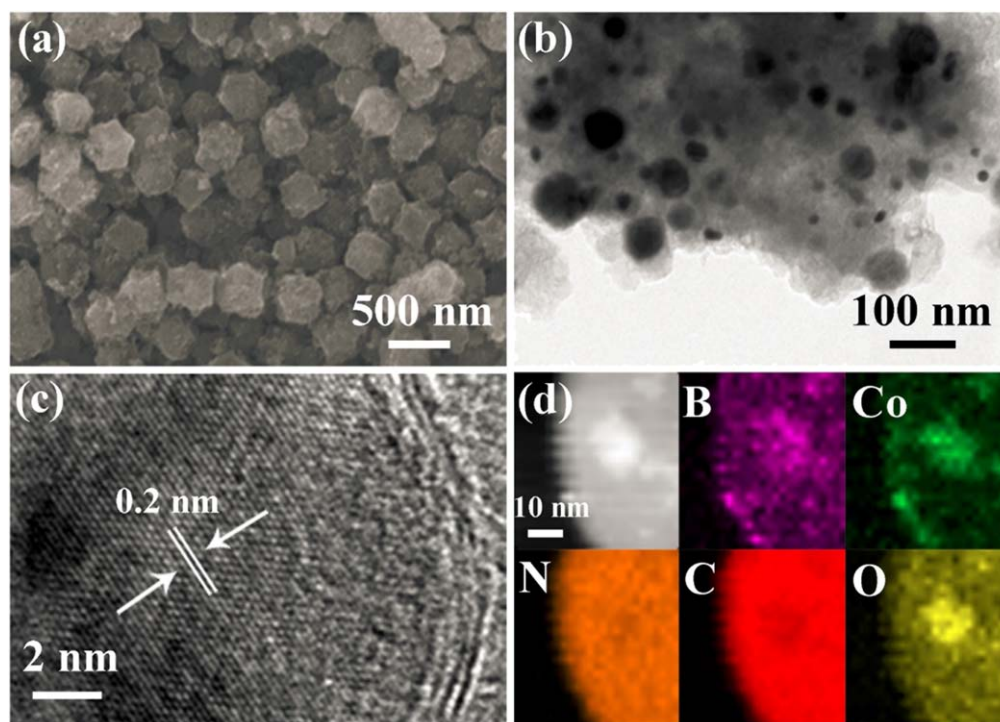


Figure 2. (a) SEM image, (b) TEM image, (c) enlarged high-resolution TEM image and (d) mapping images of B-CoNC.

discharge power density of 162.3 mW cm^{-2} and higher durability of liquid zinc-air batteries. In consideration of high safety and flexibility of solid-state batteries, we assembled a kind of all-solid-state zinc-air battery adopting gel electrolyte to illumine a light emitting diode (LED) wristband. It opens up the application in the field of solid-state batteries in terms of zinc-air batteries. On the whole, the resulting sample offers prospects for the next-generation energy system, particularly rechargeable zinc-air batteries.

Results and discussion

The CoNC sample was fabricated from the ZIFs precursors containing Co and Zn salts (figure is available online at stacks.iop.org/NANO/31/195403/mmedia) [17, 18]. On the basis of it, boron was brought in as additional dopants by pyrolyzing the mixture of as-prepared CoNC and boron acid. To eliminate the existence of B–O bonding in ultimate products, the mixture of argon and hydrogen was used as shielding gas in the carbonization process and thus, pure B-doped CoNC was obtained (denoted as B-CoNC). The introduction of B exactly modulated the distribution of adjacent electrons, thus enhancing the ORR activity [19]. For comparison, B–O contained B-doped CoNC was also prepared with only argon as the shielding gas (denoted as BO-CoNC) (figure S2). The scanning electron microscopy (SEM) image indicated B-CoNC nanoparticles maintained the polyhedral size with well-dispersed small nanoparticles even after carbonization (figure 2(a)). In the transmission electron microscopy (TEM) image of B-CoNC, dispersed small Co nanoparticles could be observed in the interior of carbon

matrix, indicating the uniform distribution of active sites and stable structure (figure 2(b)). The corresponding highly magnified TEM image (figure 2(c)) demonstrates that B-CoNC nanoparticles possess an excellent crystalline structure with a crystal lattice of 0.2 nm and is thus an indication of crystalline cobalt. High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) is then utilized to probe the elemental distribution of B-CoNC, as shown in figure 2(d). We could observe from the images that boron (pink), cobalt (green) and oxygen (yellow) mainly concentrate in the cobalt nanoparticle, while nitrogen (orange) and carbon (red) disperse throughout the substrate entirely.

The crystalline structure of three sorts of carbon materials is explored by powder x-ray diffraction (XRD). According to figure 3(a), location of $\sim 26^\circ$ exists a peak representing C (002) plane. The C (002) peak of B-CoNC reveals a spiculate peak compared with a broad and plat heave of CoNC, illustrating the incremental graphitization degree of B-CoNC. Cobalt elements have two types of crystal orientations, namely Co (111) and Co (200), locating at $\sim 44^\circ$ and $\sim 52^\circ$, respectively. The more evident peak between them is Co (111) plane, illuminating the majority form of cobalt atoms. The graphitization of carbon materials can be further studied by Raman spectroscopy (figure S3). The ratio of the D-band and G-band intensities (I_D/I_G) [20] for CoNC is 0.94. In comparison, B-CoNC and BO-CoNC nanoparticles present a relatively smaller value of 0.90, indicating more crystallized graphitic structure with heteroatom doping.

X-ray photoelectron spectroscopy (XPS) was analyzed to identify the surface element composition and bonding configuration of materials (table S1). As shown in figure 3(b), the B 1s band can be solely deconvoluted into three bands at

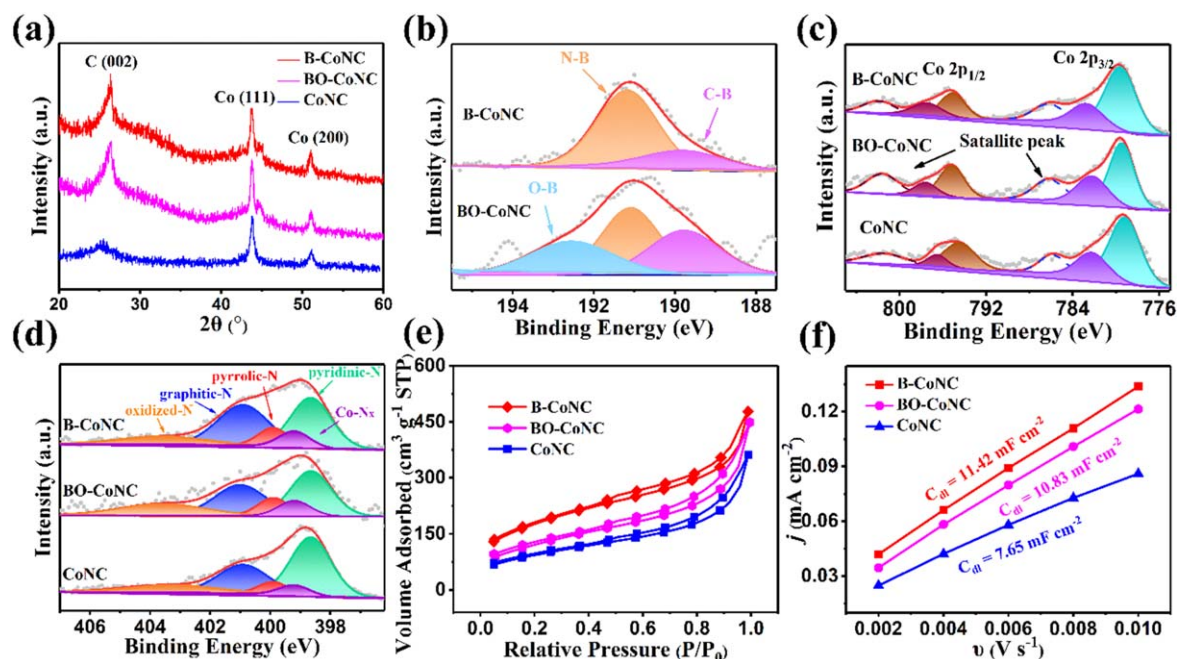


Figure 3. (a) XRD patterns, (b) B 1s XPS spectra, (c) Co 2p XPS spectra, (d) N 1s XPS spectra, (e) N₂ sorption isotherms and (f) double-layer capacitance for B-CoNC, BO-CoNC and CoNC.

189.8, 191.2, and 192.6 eV, assigned to the B–C, B–N and B–O bonding structures, respectively [21]. Specifically, while BO-CoNC possesses certain amount of B–O bonding, the elimination of B–O structure is observed in B-CoNC. As for cobalt, Co²⁺ (782.5, 797.4 eV) and Co³⁺ (778.8, 794.7 eV), namely, nitrides or oxides, are both observed in the Co 2p spectrum (figure 3(c)) [22]. Co–N_x active sites are reported to enhance the ORR process, while Co³⁺ is actually OER-related for its adsorption of intermediates in the OER process [23]. Nitrogen elements, doped in carbon materials, not only elevate the Fermi level towards conduction band but also develop a synergistic effect between nitrogen and other doped elements, thus facilitating the catalytic activity [24]. The high-resolution N 1s spectrum in figure 3(d) exhibits five types of nitrogen including pyridinic-N, graphitic-N, pyrrolic-N, Co–N_x, and oxidized-N locating at ~398.7, ~401.0, ~399.9, ~399.3 and ~403.5 eV, respectively. The graphitic-N is one of the main existence forms, while pyridinic-N and pyrrolic-N, in view of their lone-pair electrons, could integrate with cobalt to compose Co–N_x active sites [25]. It is reported that all of these N species play a crucial role in the ORR process apart from oxidized-N [26]. In fact, the electronegativity difference between boron, nitrogen and carbon atoms has been demonstrated to show a synergistic contribution to the catalytic activity of active site [27]. To verify the exposure level of active sites, the specific surface area was analyzed. As is illustrated in figure 3(e), the B-CoNC exhibits a much larger surface area of 626.42 m² g⁻¹ than BO-CoNC and CoNC, which are only 438.73 and 335.58 m² g⁻¹. The macro specific surface area indeed contributes to more exposed active sites, and sequentially improves catalytic performance. The electrochemical double-layer capacitance (denoted as C_{dl}) is deemed to reflect the catalyzing electrochemical

surface area. Cyclic voltammetry under different scanning rates was employed to measure C_{dl} of three samples (figure S4). Figure 3(f) illustrates C_{dl} of B-CoNC is 11.42 mF cm⁻² while BO-CoNC shows a comparatively low C_{dl} of 10.83 mF cm⁻² and CoNC possesses a much lower value of 7.65 mF cm⁻², which verify the enlarged surface area allows more electrocatalysts to cohere with active sites [28].

With regard to electrochemical tests, cyclic voltammetry (CV) was given in Ar-saturated 0.1 M potassium hydroxide firstly. As is indicated in figure 4(a), there is not any obvious peak in ORR process. In contrast, if it was tested in O₂-saturated KOH solution, a well-defined peak could be observed around 0.77 V versus RHE for B-CoNC, implying a superior performance of oxygen reduction activity compared with 0.76 V versus RHE for BO-CoNC and 0.75 V versus RHE for CoNC. Linear sweep voltammetry (LSV) measurements were then performed in KOH solution saturated with oxygen. Figures S5–7 portray the LSV curves under different rotating speeds and corresponding K-L plots for three electrocatalysts. According to figure 4(b), it is worth noting that CoNC catalysts exhibits inferior performance with onset potential of 0.94 V versus RHE, half-wave potential of 0.82 V versus RHE and limiting current density of 4.51 mA cm⁻², which is no match for commercial 20% Pt/C catalyst (onset potential of 0.97 V versus RHE, half-wave potential of 0.84 V versus RHE and limiting current density of 5.52 mA cm⁻²). Meanwhile, in the same circumstances, BO-CoNC material was tested, and data indicate that it has analogous performance with 20% Pt/C, which is undesirable as ever (onset potential of 0.96 V versus RHE, half-wave potential of 0.84 V versus RHE and limiting current density of 5.11 mA cm⁻²). As expected, B-CoNC distinctly had the best activity as for onset potential of 0.97 V versus RHE, half-wave potential of

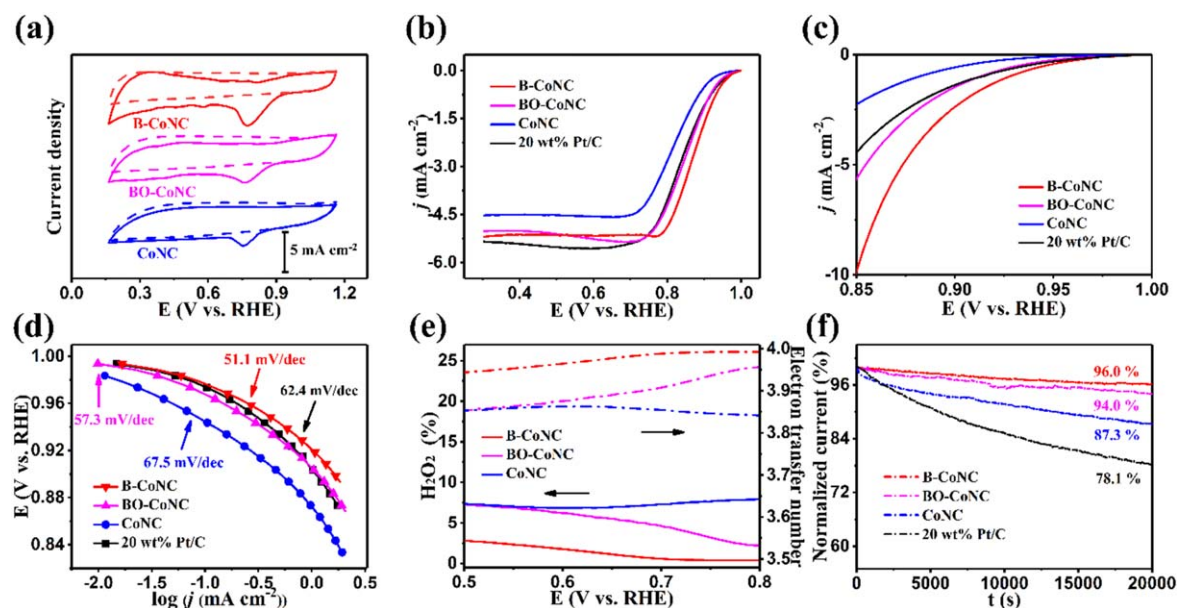


Figure 4. (a) CV curves for B-CoNC, BO-CoNC and CoNC in Ar/O₂-saturated 0.1 M KOH at room temperature with a sweep rate of 50 mV s⁻¹. (b) ORR polarization curves, (c) the kinetic current density, and (d) Tafel plots of different samples in O₂-saturated 0.1 M KOH at room temperature with a sweep rate of 10 mV s⁻¹ at a rotation speed of 1600 rpm. (e) The productivity of peroxide and electron transfer numbers of B-CoNC, BO-CoNC and CoNC based on the rotating ring disk electrode (RRDE) data. (f) ORR current-time chronoamperometric response of different samples in 0.1 M KOH solution.

0.88 V versus RHE and limiting current density of 5.21 mA cm⁻². Moreover, the kinetic current density (j_k) of B-CoNC is 9.8 mA cm⁻² at 0.85 V versus RHE, which is more than twice that of the Pt/C catalyst (4.4 mA cm⁻² at 0.85 V versus RHE) (figure 4(c)). It underlines the significance of wiping off B–O bonding by controlling experimental conditions, by which means pure B doping indeed works. Moreover, another important feature verifying predominant oxygen catalytic activity is Tafel slope that demonstrates lower overpotential of B-CoNC (51.1 mV per decade) than BO-CoNC, CoNC, and Pt/C (57.3, 67.5, 62.4 mV per decade) (figure 4(d)). Such outstanding ORR property of B-CoNC is firstly attributed to the fact that the Co active sites uniformly distributed on the surface of open carbon framework. Furthermore, B-doping into carbon materials results in a break of electroneutralities because of lower electronegativity of B (2.04) than C (2.55). In this case, positively charged boron atoms generate asymmetric spin density over active sites, further boosting the ORR activity in alkaline solution [29]. Whereas, oxygen atoms in B–O bonding influence asymmetric spin density, and only by removing oxygen could performance be optimized.

Electrochemical impedance spectroscopy (EIS) measurements were then conducted and figure S8 portrays the corresponding Nyquist plots and it is easy to discover that B-CoNC exhibits a smaller semicircle diameter than CoNC, accounting for its smaller charge transfer resistance. The increased degree of graphitization and coupling effects after doping could explain such decreasing resistance, which acts in altitudinal accordance with the LSV and Tafel data. Electron transfer number per oxygen molecule also assesses oxygen reduction reaction activity. Rotating ring disk electrode (RRDE) was utilized to calculate the electron transfer number

(n) and the productivity of intermediates (H₂O₂). As shown in figure 4(e), n of B-CoNC is about 3.99, approaching theoretical value (4) of Pt/C [30]. In addition, the chronoamperometric tests were conducted in alkaline solution to measure the durability of materials. It is presented in figure 4(f) that B-CoNC retains 96% of current density after operating 20 000 s, while BO-CoNC is slightly inferior to it, followed by undoped materials. Pt/C catalyst, however, loses almost 20% of current density if experiencing the same time and voltage [31], proving the better stability of B-CoNC.

To illustrate the bifunctional nature of catalysts, LSVs were again given in KOH aqueous solution to evaluate the OER activity (figure 5(a)). At 10 mA cm⁻², B-CoNC reaches the potential of 1.64 V versus RHE, approaching that of IrO₂ (1.61 V versus RHE) catalyst under the same condition. On the other hand, potential of CoNC to produce 10 mA cm⁻² is 1.83 V versus RHE, generating a great gap with B-CoNC, BO-CoNC (1.68 V versus RHE) and IrO₂. Figure 5(b) depicts the Tafel slope of B-CoNC (221.8 mV dec⁻¹), BO-CoNC (246.2 mV dec⁻¹), CoNC (311.4 mV dec⁻¹) and IrO₂ (109.4 mV dec⁻¹), again illustrating the desirable reaction kinetics during the OER process and further confirming the satisfactory OER activity of B-CoNC [32, 33]. The OER durability of the material was as then tested (figure 5(c)) and the current only dropped 8.5% within 20000 s, illuminating its favorable durability compared with commercial IrO₂ during persistent OER process. Figure 5(d) portrays the overall electrocatalytic property of B-CoNC, BO-CoNC, CoNC and Pt/C + IrO₂, including oxygen reduction and evolution performance. The ORR potential is taken at half-wave during oxygen reduction reaction while the OER potential is taken at a current density of 10 mA cm⁻². Thus, we define the over-voltage between ORR and OER as $\Delta E = E_{j=10} - E_{1/2}$ [34]. It

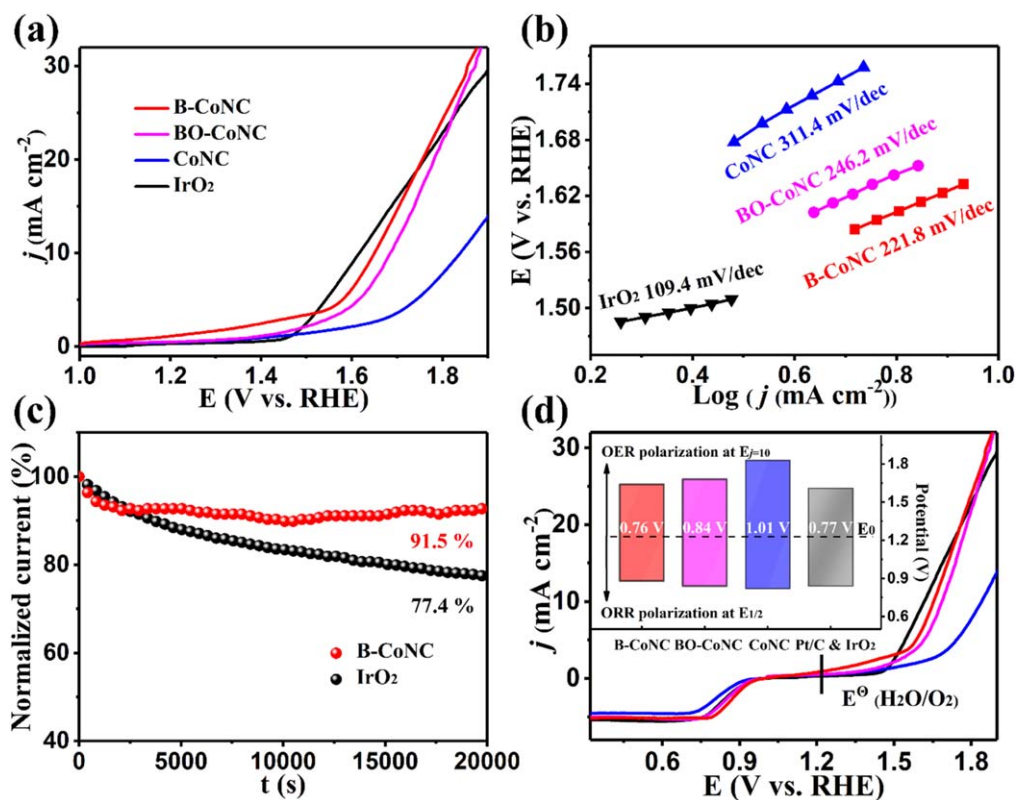


Figure 5. (a) OER polarization curves and (b) Tafel plots of B-CoNC, BO-CoNC, CoNC and IrO₂ in O₂-saturated 0.1 M KOH at a rotation speed of 1600 rpm. (c) OER durability of B-CoNC in O₂-saturated 0.1 M KOH solution. (d) The overall LSV curves of different samples (inset: Overpotential between the $E_{1/2}$ of the ORR and $E_{j=10}$ of the OER for all samples).

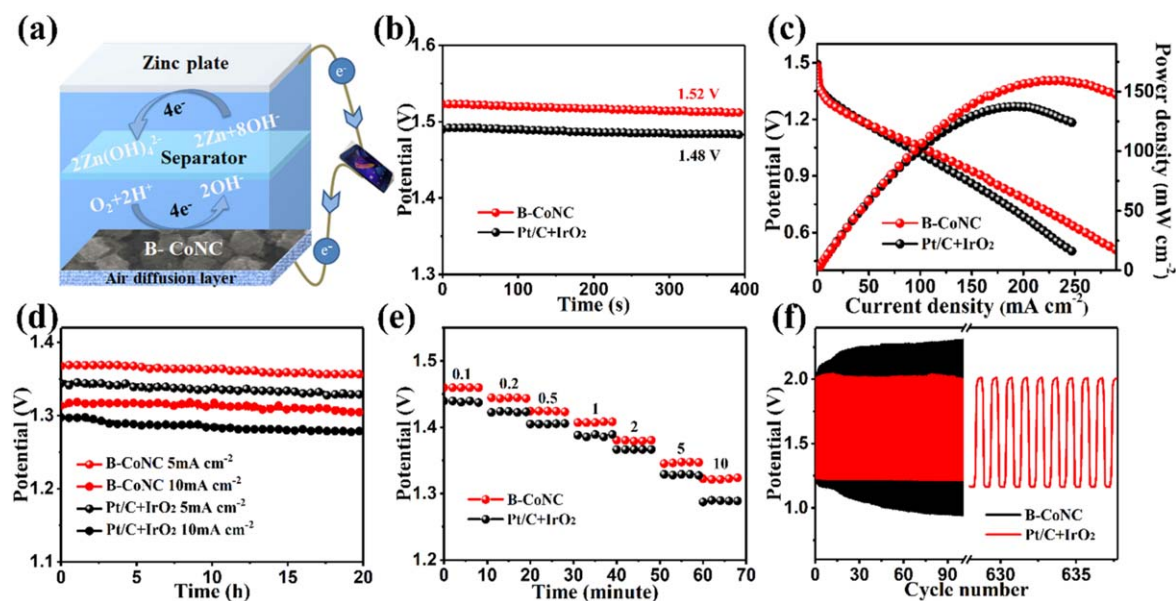


Figure 6. (a) Schematic representation of a rechargeable zinc–air battery. (b) Open-circuit plots of zinc-air batteries with two as-prepared samples as cathodes in aqueous solution. (c) Discharge voltage curves (v–i) and the corresponding power density plots using B-CoNC and Pt/C + IrO₂ air cathode. (d) Discharge curves of the primary zinc-air batteries using B-CoNC and Pt/C + IrO₂ air cathode at 5 and 10 mA cm⁻². (e) Rate discharge curves of the primary zinc-air batteries using different samples and electrolytes at different current densities. (The unit of current density is mA cm⁻².) (f) Discharge/charge cycling curves of two-electrode rechargeable zinc-air batteries using the B-CoNC and Pt/C + IrO₂ air electrode at a current density of 25 mA cm⁻².

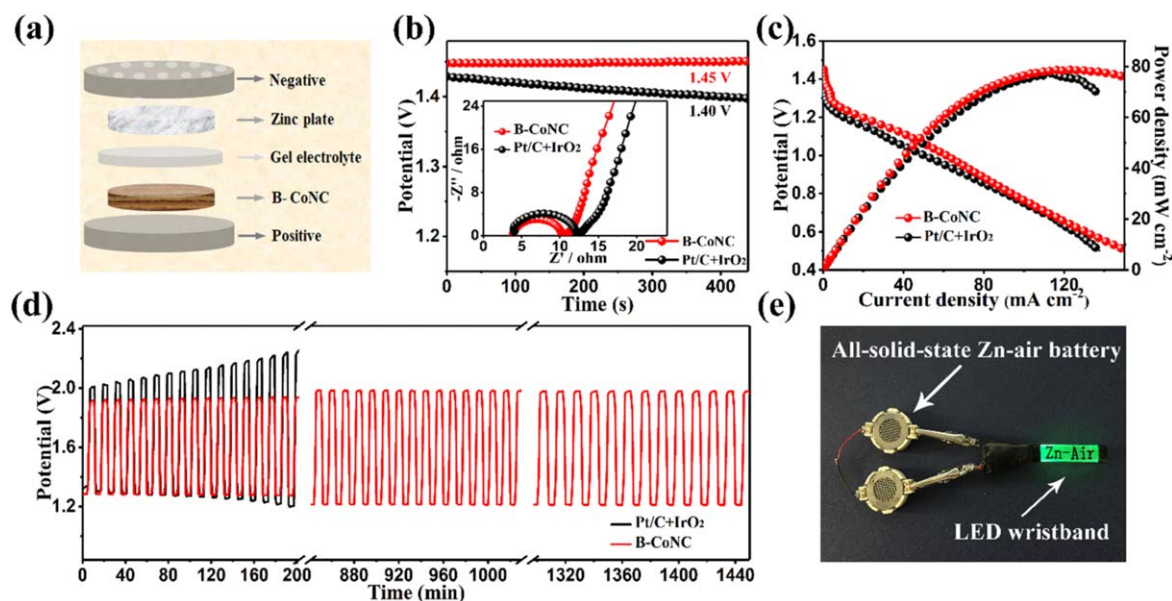


Figure 7. (a) The diagrammatic sketch of an assembled all-solid-state zinc-air battery. (b) Open-circuit plots (inset: EIS spectra), (c) discharging polarization curves and the corresponding power density plots and (d) cycling curves for B-CoNC and Pt/C + IrO₂ based all-solid-state zinc-air batteries. (e) Photograph of an LED wristband powered by two all-solid-state batteries connected in series.

is obvious in the inset graph of figure 5(d) that B-CoNC delivers a ΔE of 0.76 V that even outperformed noble metal electrocatalysts like Pt/C + IrO₂ (0.77 V), let alone BO-CoNC (0.84 V) and CoNC (1.01 V). The results are also top-ranking among previously reported works (table S2). All these suggested that our synthesized B-CoNC materials possess great potential to be applied to practical zinc-air batteries.

Driven by the increasing demand of portable electronic devices, a primary aqueous zinc-air battery was fabricated to test firstly. Carbon fiber paper coated with B-CoNC and a polished Zn plate are the cathode and anode, respectively (figure 6(a)). The open circuit voltage of the assembled zinc-air battery is 1.52 V on average in KOH electrolyte, surpassing that of Pt/C + IrO₂ (1.48 V) (figure 6(b)). The discharge voltage curve is presented in figure 6(c), and on the basis of it, the power density plot can be attained. It is worth noting that the peak power density achieves 162.3 mW cm^{-2} under the circumstance of current density attaining 226 mA cm^{-2} , which is 25 mW cm^{-2} more than Pt/C + IrO₂ electrode battery. More activity-elevated sites and porous structures facilitate the spread of oxygen and electrolyte, which can explain the excellent performance of it. When the battery was discharged at constant current density (5 and 10 mA cm^{-2}), it could hold comparatively high voltage even after 20 h (figure 6(d)). Through adjusting the current density from 0.1 to 10 mA cm^{-2} , discharge rate performance could be manifested in virtue of the decreasing discharging potential plateaus (figure 6(e)). It is transparent to find that the voltage platform of the B-CoNC based battery descends gently whereas the battery with the Pt/C + IrO₂ air electrode drops acutely in potential. A rechargeable battery with 6 M KOH and 0.2 M Zn(CH₃COO)₂ was assembled to examine the charging-discharging properties of B-CoNC-based zinc-air batteries. Low charging voltage and

high discharging voltage are required for a rechargeable battery to ensure the durability of an active catalyst. As is depicted in figure 6(f), at high rate (25 mA cm^{-2}), the zinc air battery reaches a charging plateau of 2.03 V and discharging plateau of 1.22 V, forming a comparatively small potential gap of 0.81 V. The final potentials after 635 cycles (about 127 h) are 1.97 V and 1.16 V, illustrating no apparent decline of voltage gap compared with Pt/C + IrO₂ cathode that experiences a noticeable deterioration since the first cycle. From what has been discussed above, we may safely draw a conclusion that this air cathode catalyst is comparable to some other recently reported electrocatalysts in zinc-air batteries (table S3).

Prompted by the requirement for wearable electronic devices, we fabricated an all-solid-state device with alkaline polyvinyl alcohol (PVA) as the gel electrolyte [35, 36]. Figure 7(a) presents the diagrammatic sketch of the all-solid-state structure. The OCV of the as-fabricated device attached with B-CoNC cathode catalysts keeps steady at 1.45 V (figure 7(b)). Weaker ionic conductivity of gel electrolyte is the main reason for decline in voltage between all-solid-state batteries and liquid batteries [21]. What's more, the battery with B-CoNC electrode demonstrates a slightly smaller internal resistance compared with that of Pt/C + IrO₂, illuminating a more appropriate ion channel in B-CoNC-based solid-state batteries (inset of figure 7(b)). Figure 7(c) shows the discharge voltage curves and the power density curves of all-solid-state batteries, of which peak power density came up to nearly 80 mW cm^{-2} at 120 mA cm^{-2} . When cycled at 5 mA cm^{-2} , an initial charge voltage of 1.92 V and discharge voltage of 1.28 V produces a petty voltage gap of 0.64 V and a round-trip efficiency of 66.7%. Even if after 1450 min, the battery based with B-CoNC air cathode only increases 0.13 V concerning voltage gap, certifying superb stability of such all-solid-state zinc-air batteries (figure 7(d)), which ranks at the

top level among the recently reported works (table S4). As is shown in figure 7(e), two series-connected flexible zinc air batteries equipped with B-CoNC were applied to power an LED wristband, which realized their practical application as energy devices of high safety and flexibility.

Conclusion

To sum up, the elimination of the undesired B-O bonding in B-doped catalysts could indeed maximize the heteroatom's contribution improving the oxygen reaction activity of the pristine active center, and thus boosting the cathodic reactions of zinc-air battery. Upon additional B doping, the ORR/OER bifunctional activity of B-CoNC can be further enhanced with a superior half-wave potential of 0.88 V versus RHE for ORR, a small overpotential of 410 mV at the current density of 10 mA cm⁻² for OER, and a reversible oxygen electrode index of 0.76 V in electrochemical measurements. Moreover, aqueous and all-solid-state zinc-air batteries assembled with the B-CoNC electrocatalyst manifest superior characteristics, particularly high power density and long durability. This work will open up an approach to heteroatom-doped porous carbon materials replacing noble metals for reversible energy storage and conversion devices, especially all-solid-state metal-air batteries.

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Conflict of interest

The authors declare no conflict of interest.

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